Loan Portfolio Credit Risk Analysis

Analyst’s Technical Report

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Portfolio risk profile · drivers of default · high-risk segments · default prediction · credit-risk monitoring

Item	Detail
Primary tool	Microsoft Excel (workbook: Credit_Risk_Monitoring_Tool.xlsx)
Supporting tools	Python (pandas, statsmodels, scikit-learn) for cleaning rules and model-parameter estimation; Power Query M script for Excel-native refresh
Data	credit_risk_dataset.csv — 32,581 raw loans, 32,411 after cleaning
Report date	November 2026

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1. Project overview

1.1 Project title

Loan Portfolio Credit Risk Analysis & Default Monitoring Tool - an Excel-based analytical solution for a lending institution.

1.2 Business scenario

A lending institution has asked its data team to explain the risk in its loan book. It holds historical loan applications, borrower profiles and repayment outcomes. Management wants to understand the portfolio, identify the characteristics that go with default, and improve its ability to monitor credit risk.

1.3 Business problem

About one in five loans in the book defaulted (21.9%), but the institution had no structured view of where that risk sits. Credit decisions appear to rely on the assigned loan grade, and management could not say which borrower and loan characteristics drive default, which segments are unacceptably risky, or how well default could be anticipated at application. Without that, limits and approval rules cannot be targeted, and risk cannot be monitored as the book changes.

1.4 Purpose of the analysis

To give management (credit risk, lending policy and portfolio management) an evidence base and a practical tool to: (1) measure the portfolio's risk profile; (2) identify and quantify default drivers; (3) locate high-risk segments; (4) estimate each borrower's probability of default (PD); and (5) monitor risk going forward.

1.5 Scope

- **In scope:** all 32,581 supplied records; data assessment and cleaning; descriptive, inferential and segment analysis; two predictive models with hold-out validation; an interactive Excel monitoring workbook; recommendations and monitoring indicators.
- **Out of scope (data not available):** trends over time, vintage and roll-rate analysis, loss given default and expected loss, profitability, approval/decline decisions (reject inference), and out-of-time model validation.

1.6 Tool used

Microsoft Excel is the primary tool. Every KPI, summary table, statistical test, segment ranking, model score, evaluation metric, dashboard element and finding in the workbook is a live Excel formula that recalculates when the data or settings change.

Two steps used **Python as a supporting tool**, because Excel cannot do them natively without add-ins: (1) applying the documented cleaning rules reproducibly to 32,581 records (an equivalent Power Query script is provided), and (2) estimating the model parameters - maximum-likelihood logistic regression coefficients and a CART decision tree. Those parameters were then written into the workbook and **implemented, scored and evaluated in Excel**. The logistic log-likelihood is recomputed in Excel so the coefficients can be re-verified with Solver.

2. Business objectives and analytical questions

2.1 Objectives

1. Assess the overall performance and risk profile of the loan portfolio.
2. Identify customer and loan characteristics associated with default.
3. Evaluate how loan grades and interest rates relate to default rates.
4. Identify high-risk loan segments.
5. Develop a predictive model to estimate the likelihood of loan default.
6. Translate the findings into recommendations for credit risk monitoring.

2.2 Analytical questions

- **Portfolio performance:** How many loans are represented? What percentage of borrowers defaulted? What is the total value of loans issued? What is the distribution across credit grades? Which loan purposes account for the largest share of lending?
- **Risk segmentation:** Which grades have the highest default rates? Is income associated with default? How does loan-to-income relate to default? Do shorter employment histories default differently? How does default vary across purposes and home-ownership categories? Is there a relationship between interest rates and default?
- **Predictive analysis:** Can we estimate a borrower's probability of default? Which variables are most useful? How does logistic regression compare with a tree-based model? What proportion of actual defaults can the model identify?

2.3 Objective-to-analysis mapping

Objective	Analytical question	Data required	Analysis	Output (sheet)
1 Portfolio profile	Size, default rate, value lent, concentration	loan_amnt, loan_status, grade, intent, home	COUNTIFS/SUMIFS KPIs, distribution and rank tables	Portfolio, Dashboard
2 Default drivers	Which characteristics go with default, how strongly?	All borrower and loan fields	Default-rate tables, chi-square + Cramér's V, Welch t-tests, point-biserial r	Risk_Drivers
3 Grades and rates	Do grade and rate track default? Does rate add to grade?	loan_grade, loan_int_rate, loan_status	Grade table, rate bands, within-grade median split, t-test	Grade_Rate
4 High-risk segments	Where do defaults concentrate?	grade × home × LTI band	Heat maps; ranked 140-segment table with size threshold	Segments
5 Prediction	Can PD be estimated? Best model? Recall?	Train/test split, all predictors	Logistic regression and decision tree; AUC, confusion matrix, gains, calibration; IV	Model_LR, Model_Tree, Model_Eval, Predictive_Power, Scorer
6 Recommendations	What should management do and monitor?	Outputs of 1–5	Finding → evidence → action; KRIs with current values	Insights

Gap check: every objective can be addressed with the data. The data cannot support time trends, loss estimation or out-of-time validation; these are recorded as limitations rather than attempted.

3. Dataset overview

Item	Detail
Source file	credit_risk_dataset.csv (single flat file, UTF-8, comma-separated)
Records / fields	32,581 records × 12 fields (one row per loan)
Identifiers	None supplied — loan_id created from file order
Dates	None — no application, disbursement or default dates
Outcome	loan_status (1 = default, 0 = no default); 21.8% default in raw file
Numeric fields	person_age, person_income, person_emp_length, loan_amnt, loan_int_rate, loan_percent_income, cb_person_cred_hist_length
Categorical fields	person_home_ownership (4), loan_intent (6), loan_grade (7), cb_person_default_on_file (Y/N)
Relationships	Single table; no joins or keys required
Provenance	Undocumented; matches a widely circulated public practice dataset. Treated as illustrative data, not real institutional records

3.1 Important fields

- **loan_status** — the target for every default-rate KPI and both models.
- **loan_grade** — the lender's own risk rating, the benchmark the models must beat.
- **loan_amnt, person_income → lti** — affordability; turned out to be the strongest driver.
- **person_home_ownership** — interacts strongly with affordability.
- **loan_int_rate** — pricing; tested for information beyond grade.

3.2 Dataset limitations

No dates, identifiers, loss amounts, default definition or currency; only approved loans are present; some relationships are near-deterministic. Section 10 discusses the consequences.

4. Data quality assessment

Every field was profiled for type, range, missingness, distinct values and internal consistency before any change was made.

Dimension	Finding	Impact	Action
Completeness	person_emp_length missing 895 (2.7%); loan_int_rate missing 3,116 (9.6%); all other fields complete	Employment-missing borrowers default at 31.7% (informative). Rate missingness is even across grades with an unchanged default rate (random-looking)	Retain and flag; explicit "Unknown"/"Not recorded" categories
Uniqueness	165 exact duplicate rows (all 12 fields identical). Second copies cluster in rows ~15,977–17,999 and 25,000+	Double-counting; duplicates default at 11.5%, so they bias the rate down	Remove later copy
Validity	Ages 123 and 144 (5 records, one with income 6,000,000); employment 123 years at ages 21–22 (2); rate exactly 6.00% in grades B–E (11), below each grade's otherwise-minimum rate	Distorts age, income and rate statistics	Exclude invalid-age records; set the other values to missing
Consistency	Supplied loan_percent_income ≠ loan_amnt ÷ income beyond rounding in 429 raw records (428 after cleaning) — every one a default; supplied ratio ≈ 0.85 × computed	Could bias LTI analysis and models	Recompute LTI, flag, run sensitivity check
Accuracy	Extreme but plausible incomes up to ~2.0M; 84- and 94-year-old borrowers	Skew in means	Retain; use bands and ln(income)
Formatting	No whitespace, case or spelling variants in categorical fields; numeric types parse cleanly	None	None required
Measurement	Default definition, observation window and currency undocumented; "OTHER" home ownership undefined	Limits interpretation	Documented as assumptions

A further structural observation: the file is ordered in blocks (for example, average credit-history length jumps between row ranges), so any sequential train/test split would be biased. A stratified random split was used.

5. Data cleaning and preparation

5.1 Cleaning and transformation log

#	Issue	Variable	Action	Reason	Effect
1	No unique record identifier	All records	Created loan_id = 'LN' + raw row position	Needed to trace cleaned and excluded records back to the raw file and to support audit.	32,581 IDs created before any exclusions.
2	Exact duplicate records (all 12 fields identical)	All fields	Removed the later copy, kept the first occurrence	165 pairs; second copies sit in two contiguous blocks of the file (rows ~15,977–17,999 and 25,000+), consistent with an overlapping extract rather than coincidence.	−165 records. Duplicates had an 11.5% default rate, so keeping them would slightly understate risk.
3	Impossible ages	person_age	Excluded records with age > 100	Ages of 123 and 144 are impossible; the same records include an income of 6,000,000, so the whole record is unreliable.	−5 records (all non-defaults). Ages 84 and 94 retained as plausible.
4	Impossible employment length	person_emp_length	Set to missing (record kept)	123 years of employment for borrowers aged 21–22. The rest of each record looks valid.	2 values set to missing; counted with the 'Unknown' employment group.
5	Missing employment length	person_emp_length	Retained as missing; 'Unknown' band; emp_unknown flag; median (4 yrs) imputation for the tree only	Missingness is informative (31.7% default vs 21.9% overall), so deleting or silently imputing would hide a risk signal.	889 records (2.7%) flagged.
6	Placeholder interest rates	loan_int_rate	Set 6.00% to missing for grades B–E	6.00% sits far below each grade's otherwise-minimum rate (B 8.70%, C 10.28%, D 11.86%, E 13.43%). For grade A it is plausible and kept.	11 values set to missing.
7	Missing interest rate	loan_int_rate	Retained as missing; 'Not recorded' band; excluded from rate averages and tests; grade-median imputation for the tree only	Missing evenly across grades (8–11%) with a similar default rate (20.7% vs 21.9%), consistent with random missingness. Rate is not used by the final logistic model.	3,105 records (9.6%) not recorded.
8	Supplied loan_percent_income inconsistent with its components	loan_percent_income	Recomputed lti = loan_amnt ÷ person_income; flagged LTI_MISMATCH; supplied field kept for audit	428 records differ beyond 2-dp rounding and every one is a default (supplied ratio ≈ 0.85 × computed). The recomputed value is precise and consistent with the loan amounts used for exposure.	Sensitivity check: band default rates move by ≤2.5 points and conclusions are unchanged.
9	Extreme incomes	person_income	Retained	Values up to ~2.0M are plausible; no evidence of entry error once the age-144 record is excluded.	Handled with income bands and ln(income) rather than removal.
10	Categorical label consistency	Text fields	Checked; no change required	No leading/trailing spaces, case or spelling variants found.	None.
11	Model development split	All records	Stratified 70/30 Train/Test split (seed 42)	The file is ordered in blocks (e.g. by credit history), so a random stratified split avoids a biased hold-out set.	Train 22,687 / Test 9,724; identical default rates (21.87%).
12	Analysis fields	Derived	Added lti, ln_income, grade_num, emp_unknown, imputed model inputs, 7 band fields, model scores and risk band as live table formulas	Lets every summary update automatically and lets new records be scored when added to the table.	19 formula columns in tblLoans.

5.2 Record reconciliation

Stage	Records
Raw file	32,581
Less exact duplicates	−165

Stage	Records
Less invalid ages	−5
Cleaned analysis table (tblLoans)	32,411
Training sample (70%, stratified)	22,687
Test sample (30%)	9,724

5.3 Derived variables

Field	Definition / Excel formula
lti	=[@loan_amnt]/[@person_income]
ln_income	=LN([@person_income])
grade_num	=MATCH([@loan_grade],GradeList,0) (A = 1 … G = 7)
emp_unknown	=IF([@person_emp_length]="",1,0)
emp_length_model	Missing → frozen training median (4 yrs); tree input only
int_rate_model	Missing → frozen training median for the grade; tree input only
7 band fields	=INDEX(Band_LTI_Lab,MATCH([@lti],Band_LTI_Lo,1)) etc.; "Unknown"/"Not recorded" for missing
lr_logit, lr_pd	Logistic linear predictor and =1/(1+EXP(-[@lr_logit]))
tree_leaf, tree_pd	Nested IF routing to L01–L32; leaf PD lookup
model_pd, risk_band	Champion-model PD and its band (Low / Moderate / High / Very high)

5.4 Sensitivity check on the loan-to-income inconsistency

Because every inconsistent record is a default, the choice between the supplied and recomputed ratio could tilt results. Re-running the band analysis with the supplied field moved default rates by at most 2.5 percentage points (30-40% band: 64.2% recomputed vs 61.9% supplied; renters with LTI ≥ 30%: 94.8% vs 92.3%). The logistic model's test AUC changed by 0.001 during development. No conclusion depends on the choice. The recomputed value was kept because it is exact and consistent with the loan amounts used for exposure.

5.5 Raw data preservation

The raw file is untouched in data/raw/ and on the Data_Raw sheet (table tblRaw). Excluded records, with reasons, are on Data_Exclusions and in excluded_records.csv. The cleaned table keeps the supplied loan_percent_income for audit and a dq_flags column on every record.

6. Tool-specific execution (Excel)

This section explains exactly how the work was implemented in the workbook. The file contains 21 sheets and about 620,000 formulas. It uses no macros, no external connections and no dynamic-array (spilling) functions, so it works in Excel 2016 and later.

6.1 Workbook architecture

Layer	Sheets	Role
Navigation	Home	Purpose, sheet map with hyperlinks, QA status, colour key, data caveats
Presentation	Dashboard, Insights	Interactive monitoring view; live findings and KRIs
Analysis	Portfolio, Risk_Drivers, Grade_Rate, Segments, Predictive_Power	Objectives 1–4 and variable usefulness
Modelling	Model_LR, Model_Tree, Model_Eval, Scorer	Objective 5 and the applicant tool
Control	QA_Checks, Lists, Dashboard_Calc	Reconciliations; settings, bands and lists; chart helper tables
Data	Data_Clean (tblLoans), Data_Raw (tblRaw), Data_Exclusions (tblExcluded)	Analysis table with formula columns; raw and excluded records
Documentation	Cleaning_Log, Data_Dictionary, KPI_Definitions	In-workbook documentation

Inputs are visually separated from formulas: blue text on yellow cells are editable settings (filters, thresholds, Scorer inputs, model coefficients); black cells are formulas; grey cells are fixed estimation outputs.

6.2 Structured table and calculated columns

What: the cleaned data sits in the Excel table tblLoans (32,411 rows × 34 columns): 15 value columns and 19 calculated columns. **Why:** structured references make every summary formula readable (tblLoans[loan_status]) and make the tool extensible — new rows pasted under the table are banded and scored automatically and all summaries include them without editing ranges. **How:** calculated-column formulas use this-row references, for example:

```
lti        =[@loan_amnt]/[@person_income]
lti_band   =INDEX(Band_LTI_Lab,MATCH([@lti],Band_LTI_Lo,1))
emp_band   =IF([@person_emp_length]="","Unknown",INDEX(Band_Emp_Lab,MATCH([@person_emp_length],Band_Emp_Lo,1)))
```

Output: analysis-ready fields used by every other sheet. Band thresholds live on the Lists sheet, so management can change a cut-off and the whole workbook re-bands. Approximate-match INDEX/MATCH was chosen over XLOOKUP for compatibility with Excel 2016/2019.

6.3 Portfolio KPIs and distribution tables (Objective 1)

Headline KPIs use whole-table functions (ROWS, SUM, AVERAGE, MEDIAN) and conditional aggregation; each is stored in a defined name (KPI_Loans, KPI_DR, KPI_Value …) so other sheets can refer to it by meaning rather than address. Distribution tables combine:

```
Loans          =COUNTIFS(tblLoans[loan_grade],"A")
Value lent     =SUMIFS(tblLoans[loan_amnt],tblLoans[loan_grade],"A")
Avg rate       =IFERROR(AVERAGEIFS(tblLoans[loan_int_rate],tblLoans[loan_grade],"A"),0)
Cumulative %   =SUM($D$24:D24)          Rank by value  =RANK.EQ(F24,$F$24:$F$30,0)
```

Totals rows reconcile to the headline KPIs (checked on QA_Checks). Data bars show share; a green–red colour scale highlights default rates.

6.4 Statistical testing in Excel (Objectives 2 and 3)

Chi-square test of independence for each banded characteristic against default. Rather than building separate expected-count tables, the statistic is computed directly from observed counts with SUMPRODUCT, where p is the overall default rate:

```
χ²  =SUMPRODUCT((Defaults-Loans*p)^2/(Loans*p))+SUMPRODUCT((NonDefaults-Loans*(1-p))^2/(Loans*(1-p)))
p-value =CHISQ.DIST.RT(χ², rows-1)      Cramér's V =SQRT(χ²/N)
```

An assumption check reports the smallest expected count (MIN(Loans)*MIN(p,1-p)), which is ≥ 5 for every test. Because p-values are vanishingly small at n = 32,411, effect size (Cramér's V) is used to rank practical importance.

Welch's t-test compares defaulters and non-defaulters on continuous measures without assuming equal variances. Group standard deviations come from SUMPRODUCT with a mask that ignores blanks:

```
SD(default) =SQRT(SUMPRODUCT((tblLoans[loan_status]=1)*(x<>"")*(x-mean1)^2)/(n1-1))
t =(m1-m0)/SQRT(s1^2/n1+s0^2/n0)   df = Welch–Satterthwaite   p =T.DIST.2T(ABS(t),df)
```

Cohen's d and the point-biserial correlation (=(m1-m0)*SQRT(n1*n0)/((n1+n0)*STDEV.P(x))) give effect sizes. Income is tested on ln(income) because of its skew. Grade medians use classic array formulas such as {=MEDIAN(IF((tblLoans[loan_grade]="A")*(tblLoans[loan_int_rate]<>""),tblLoans[loan_int_rate]))}; MINIFS/MAXIFS give rate ranges. The within-grade test splits each grade at its median rate with COUNTIFS(...,"<="&median).

6.5 Weight of Evidence and Information Value

On the training rows only, each band's share of non-defaults and defaults (+0.5 adjustment) gives WoE =LN(%good/%bad) and IV =Σ(%good−%bad)×WoE. A summary table ranks IV beside the tree's variable importance and whether each variable entered the logistic model — answering "which variables are most useful".

6.6 Segmentation (Objective 4)

Three heat maps (grade × LTI, housing × LTI, purpose × housing) show default rate beside the matching loan counts, so small cells are not over-read. A 140-row table enumerates every grade × housing × LTI segment with loans, defaults, rate, value, lift and share of defaults. Ranking honours a user-set minimum size through a sort key and returns the Top 15 with LARGE + INDEX/MATCH:

```
Rank key  =IF(Loans>=Seg_MinN, ROUND(rate,6)*1000000 + Loans/100000 + ROW()/1000000000, -1)
Segment k =IF(LARGE(keys,k)<0,"",INDEX(segments,MATCH(LARGE(keys,k),keys,0)))
```

The key breaks ties by size and position so the list is deterministic. Changing the minimum size or the high-risk threshold updates the ranking, the flags and the summary of flagged segments.

6.7 Logistic regression implementation (Objective 5)

Coefficients sit in a parameter table (variable, level, coefficient) including zero rows for reference categories. The model is scored in tblLoans with SUMIFS lookups for categorical effects and defined names for continuous ones:

```
lr_logit =LR_b0 + SUMIFS(LR_Coef,LR_Var,"loan_grade",LR_Level,[@loan_grade])
        + SUMIFS(...,"person_home_ownership",...) + SUMIFS(...,"loan_intent",...)
        + LR_b_LTI*[@lti] + LR_b_LTI2*[@lti]^2 + LR_b_LnInc*[@ln_income] + LR_b_EmpUnk*[@emp_unknown]
lr_pd    =1/(1+EXP(-[@lr_logit]))
```

The sheet derives z-scores, p-values (NORM.S.DIST), odds ratios and 95% confidence intervals from the coefficients and standard errors, and recomputes model fit in Excel: log-likelihood =SUMPRODUCT((split="Train")*(y*LN(pd)+(1-y)*LN(1-pd))), null log-likelihood, likelihood-ratio χ^2, McFadden R^2 and AIC. Because the mean predicted PD on the training rows equals the actual default rate (a property of the maximum-likelihood solution), QA_Checks can confirm the coefficients are at the optimum; Solver instructions let a reviewer re-maximise the log-likelihood in Excel.

6.8 Decision tree implementation

The 32-leaf tree is implemented as one nested IF formula (six levels deep) in tblLoans[tree_leaf], for example:

```
=IF([@lti]<=0.304417, IF([@grade_num]<=3.5, IF([@person_income]<=19996, ... ,"L04"), ...),
  IF([@person_home_ownership]<>"RENT", ... , IF([@lti]<=0.307851,"L31","L32")))
```

Thresholds were rounded for readability and then checked: all 32,411 loans route to exactly the same leaf as the original model. The Model_Tree sheet lists each rule in plain English with live train and test counts (COUNTIFS on leaf and split) and the leaf PD, =(defaults+1)/(loans+2) (Laplace correction, so no rule predicts exactly 0% or 100%).

6.9 Model evaluation

All evaluation uses the 9,724 test loans. Confusion matrices are COUNTIFS on split, outcome and pd>=threshold, with the threshold and a cost ratio as inputs. A 101-row ROC table gives true- and false-positive rates at thresholds 0.00–1.00; AUC is the trapezoid sum, KS the maximum TPR − FPR, and the lowest-cost threshold =INDEX(thresholds,MATCH(MIN(cost),cost,0)) with cost = missed defaults × ratio + false alarms. A grade-only ROC (flag grade ≥ k) provides the baseline. A gains table uses {=LARGE(IF(split="Test",pd),k)} to find cut-offs and reports both defaults caught and the share actually reviewed (the tree has tied PDs). A calibration table compares mean predicted PD with the actual rate by PD band. ROC and gains charts are scatter charts driven by these tables.

6.10 Interactive dashboard

Four data-validation drop-downs (grade, purpose, housing, LTI band) each include "All". A helper converts "All" to the wildcard "*", so a single COUNTIFS works for any combination:

```
crit_grade =IF(Dashboard!$C$6="All","*",Dashboard!$C$6)

Sel_N
=COUNTIFS(tblLoans[loan_grade],crit_grade,tblLoans[loan_intent],crit_intent,tblLoans[person_home_ownership],crit_home,tblLoans[lti_band],crit_lti)
```

Eight KPI cards, a dynamic subtitle and a conditionally formatted alert ("HIGH RISK", "Above average", "At or below", "Too few loans") respond instantly. Each chart's helper table applies every filter except its own dimension, so the grade chart still compares all grades within the chosen purpose and housing type. A Top-5 risk-segment panel links to the Segments sheet.

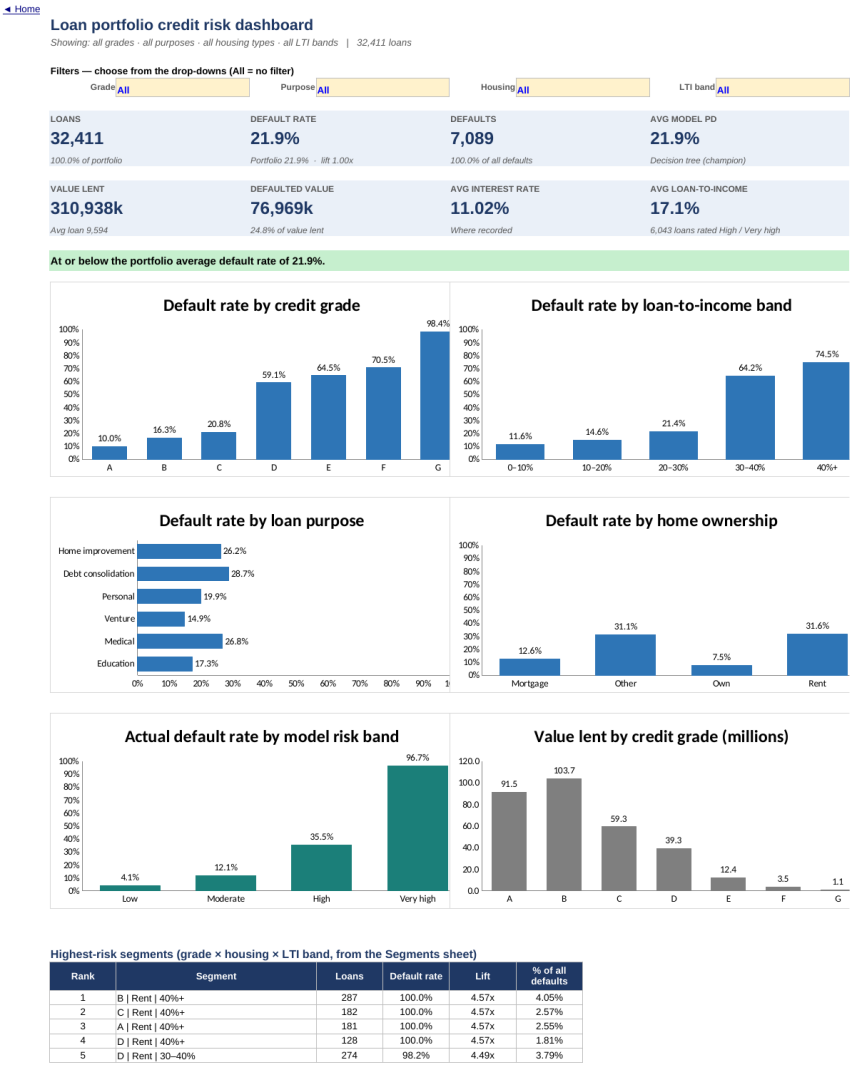


Figure 1. Dashboard (unfiltered view), exported from the workbook.

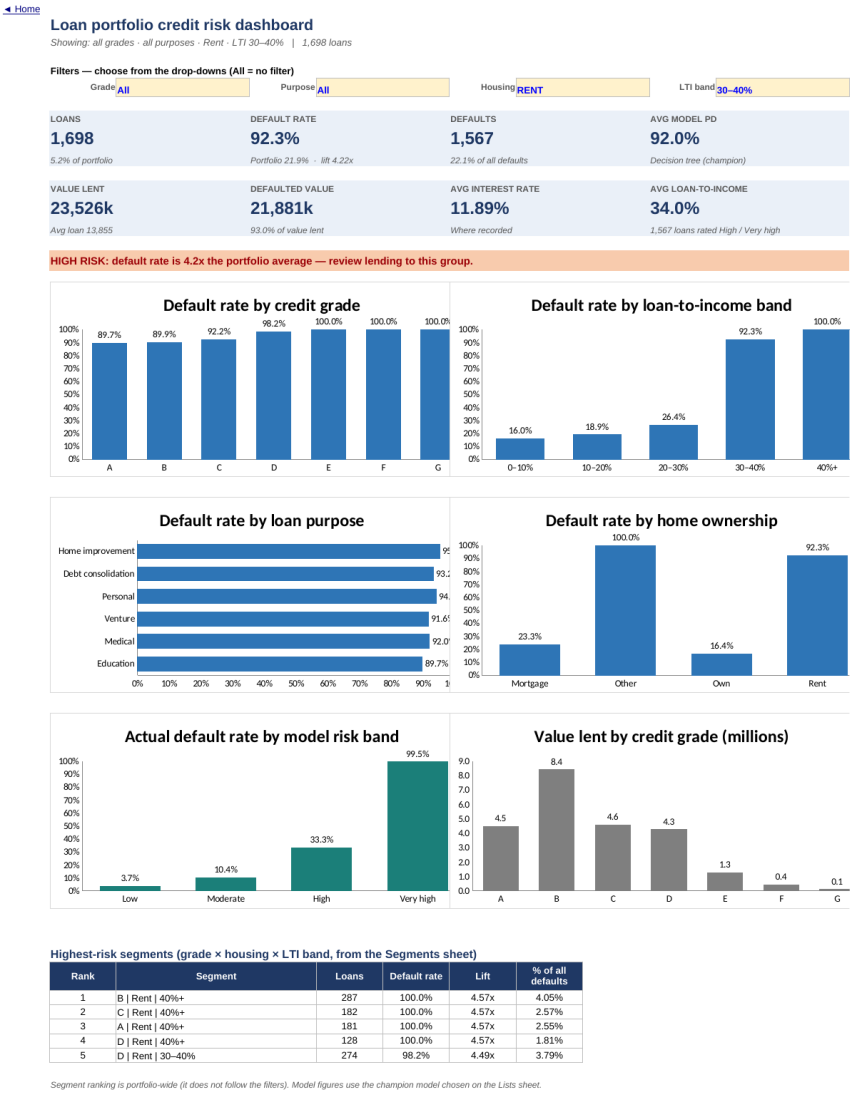


Figure 2. Dashboard filtered to renters with LTI 30–40%: the alert switches to HIGH RISK.

6.11 Applicant scorer, insights and QA

- **Scorer:** validated inputs (lists, numeric ranges, input-check message), derived inputs, both model PDs, the matching tree rule, champion PD, risk band, suggested action and the historical default rate of similar loans (same grade, housing and LTI band).
- **Insights:** ten findings whose evidence is a live sentence built with TEXT() and defined names, so the narrative updates with the data; plus monitoring indicators with current values.
- **QA_Checks:** 28 checks (record counts, exclusions, reconciliation of totals, split balance, leaf coverage, PD validity, MLE property, agreement with independent Python results, frozen-parameter governance), each with tolerance and PASS/FAIL, rolled up to a status shown on Home.

6.12 Supporting tools and why

Tool	Purpose	Why not Excel alone
pandas	Apply cleaning rules; create IDs, flags, bands and the stratified split	Reproducible, scripted rules; Power Query equivalent supplied (not run in the build environment)
statsmodels	Maximum-likelihood logistic regression; standard errors; likelihood-ratio test for variable selection	Excel has no logistic regression; Solver can confirm coefficients but not standard errors
scikit-learn	CART decision tree with 5-fold cross-validated tuning; gradient-boosting benchmark	Excel cannot grow a tree; the fitted tree is implemented as formulas

7. Objective-by-objective analysis

7.0 Direct answers to the business questions

Question	Answer
How many loans?	32,411 after cleaning (32,581 in the raw file).
What % defaulted?	21.9% (7,089 loans); 24.8% by value.
Total value issued?	310,937,900 (currency not documented); 76,968,675 in loans that defaulted.
Distribution across grades?	A 33.0%, B 32.0%, C 19.9%, D 11.2%, E 3.0%, F 0.7%, G 0.2%.
Largest purposes by lending?	Fairly even: Education 19.5%, Medical 18.0%, Venture 17.5% of value lent; home improvement smallest (12.0%).
Highest-default grades?	G 98.4%, F 70.5%, E 64.5%, D 59.1%; A–C 10.0–20.8%.
Income and default?	Yes, strong and negative: under 25k 54.6% vs 120k+ 8.4%; Welch t on ln(income) = -50.0, Cohen's d = -0.71.
Loan-to-income and default?	The strongest driver: 11.6% (<10%) → 21.4% (20–30%) → 64.2% (30–40%) → 74.5% (40%+); Cramér's V 0.432.
Shorter employment?	Yes, modestly: under 1 yr 28.0% vs over 10 yrs 16.2%; unknown 31.7%; V 0.107 (small).
Purpose and home ownership?	Debt consolidation 28.7%, medical 26.8%, home improvement 26.2% highest; venture 14.9% lowest. Renters 31.6%, mortgage 12.6%, owners 7.5%.
Interest rate and default?	Yes overall (r = 0.336; 9.4% under 8% vs 68.8% at 18%+), but it mirrors grade and adds little within a grade.
Can PD be estimated?	Yes: both models score every loan; the tree's Very-high band defaults at 96.7% and its Low band at 4.1%.
Most useful variables?	Loan-to-income, grade, housing and income; then purpose and employment-data availability. Age and credit-history length add nothing.
Logistic vs tree?	Tree better: test AUC 0.905 vs 0.880 (grade alone 0.728).
Share of defaults identified?	At a 50% threshold: tree 68.3%, logistic 57.5%. At the lowest-cost threshold (cost ratio 5): tree 74.1% with 4.1% of good loans flagged.

7.1 Objective 1 - Portfolio performance and risk profile

What it establishes: the size, outcome and concentration of the book. **Data:** loan amount, status, grade, purpose, housing. **Procedure:** headline KPIs and distribution tables (Portfolio sheet).

Grade	Loans	% loans	Value lent	% value	Avg rate %	Default rate
A	10,703	33.0%	91,464,650	29.4%	7.33	10.0%
B	10,384	32.0%	103,742,875	33.4%	11.00	16.3%
C	6,436	19.9%	59,343,250	19.1%	13.47	20.8%
D	3,620	11.2%	39,297,850	12.6%	15.37	59.1%
E	963	3.0%	12,441,875	4.0%	17.02	64.5%
F	241	0.7%	3,546,875	1.1%	18.61	70.5%
G	64	0.2%	1,100,525	0.4%	20.25	98.4%

Purpose	Loans	% loans	Value lent	% value	Rank (value)	Default rate
Education	6,409	19.8%	60,750,200	19.5%	1	17.3%
Medical	6,042	18.6%	56,018,875	18.0%	2	26.8%
Venture	5,680	17.5%	54,421,500	17.5%	3	14.9%
Personal	5,497	17.0%	52,664,100	16.9%	4	19.9%
Debt consolidation	5,189	16.0%	49,841,750	16.0%	5	28.7%
Home improvement	3,594	11.1%	37,241,475	12.0%	6	26.2%

Interpretation: the book is concentrated in the better grades (A-C hold 84.9% of loans) and spread evenly across purposes, with education the largest by value. Renters take 50.5% of loans. Average loan 9,594; average LTI 17.1%;

12.8% of loans have LTI of 30% or more. **Implication:** a 21.9% default rate is high for any lender, and the value-weighted rate (24.8%) shows larger loans default more. **Limitation:** without dates, the trend cannot be assessed.

7.2 Objective 2 - Characteristics associated with default

Procedure: default-rate tables and chi-square tests for nine banded characteristics (Risk_Drivers), plus grade and rate (Grade_Rate); Welch t-tests for continuous measures.

Characteristic	χ^2	df	p-value	Cramér's V	Effect
Loan-to-income band	6,048	4	< 1e-300	0.432	Medium
Loan grade	5,587	6	< 1e-300	0.415	Medium
Interest-rate band	4,149	7	< 1e-300	0.358	Medium
Income band	2,942	5	< 1e-300	0.301	Medium
Home ownership	1,895	3	< 1e-300	0.242	Small
Prior default on file	1,043	1	9.9e-229	0.179	Small
Loan purpose	520	5	4.8e-110	0.127	Small
Loan amount band	445	4	5.2e-95	0.117	Small
Employment length	371	5	4.7e-78	0.107	Small
Age band	30	4	4.4e-6	0.031	Negligible
Credit history length	26	4	3.6e-5	0.028	Negligible

Measure	Mean (default)	Mean (no default)	Welch t	Cohen's d	r (pb)
ln(income)	10.623	11.010	-50.0	-0.71	-0.283
Loan-to-income	0.249	0.149	60.2	1.01	0.386
Interest rate %	13.07	10.44	57.8	0.86	0.336
Employment (yrs)	4.12	4.97	-15.8	-0.21	-0.086
Age	27.47	27.80	-3.9	-0.05	-0.022
Credit history (yrs)	5.68	5.85	-2.9	-0.04	-0.016

Interpretation: affordability dominates. LTI (V 0.432, d 1.01) and income (V 0.301) are the strongest borrower-level signals; housing is moderate (renters 31.6% vs owners 7.5%); purpose and employment length are weaker but real; unknown employment (31.7%) is riskier than any known band. Age and credit-history length are statistically significant only because n is large — the effects are negligible (V ≈ 0.03, d ≈ 0.05). A prior default doubles the default rate (37.9% vs 18.4%), but the flag appears only in grades C–G. **Implication:** affordability and housing should sit beside grade in credit policy. **Limitation:** associations, not causes; income and LTI are correlated, which the models handle jointly.

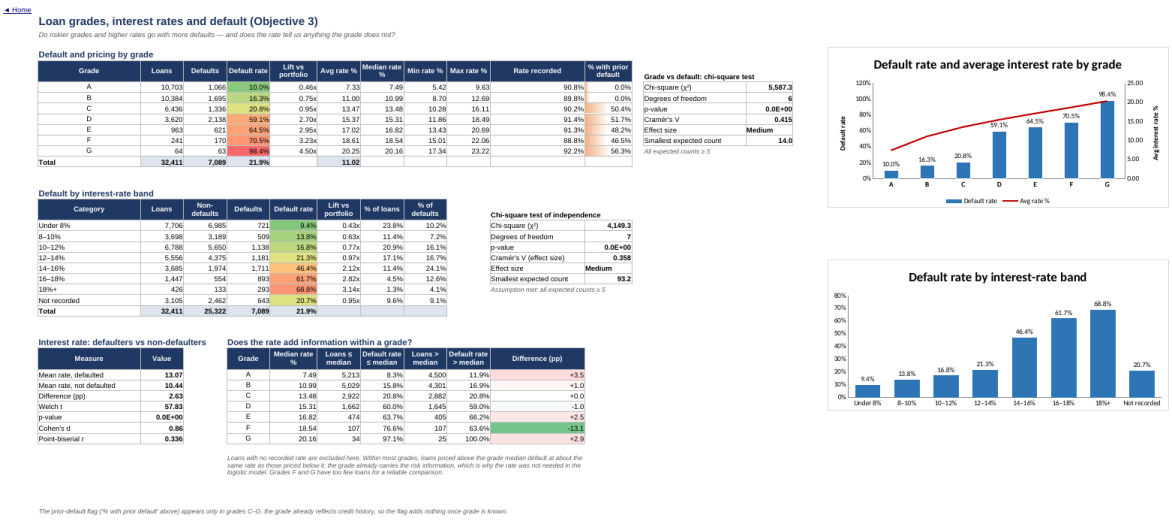


Figure 3. Grade_Rate sheet: default and pricing by grade, rate bands and the within-grade test.

7.3 Objective 3 - Loan grades, interest rates and default

Grade	Default rate	Avg rate %	Rate range %	% with prior default	Rate ≤ median: default	Rate > median: default
A	10.0%	7.33	5.42–9.63	0.0%	8.3%	11.9%
B	16.3%	11.00	8.70–12.69	0.0%	15.8%	16.9%
C	20.8%	13.47	10.28–16.11	50.4%	20.8%	20.8%
D	59.1%	15.37	11.86–18.49	51.7%	60.0%	59.0%
E	64.5%	17.02	13.43–20.69	48.2%	63.7%	66.2%
F	70.5%	18.61	15.01–22.06	46.5%	76.6%	63.6%
G	98.4%	20.25	17.34–23.22	56.3%	97.1%	100.0%

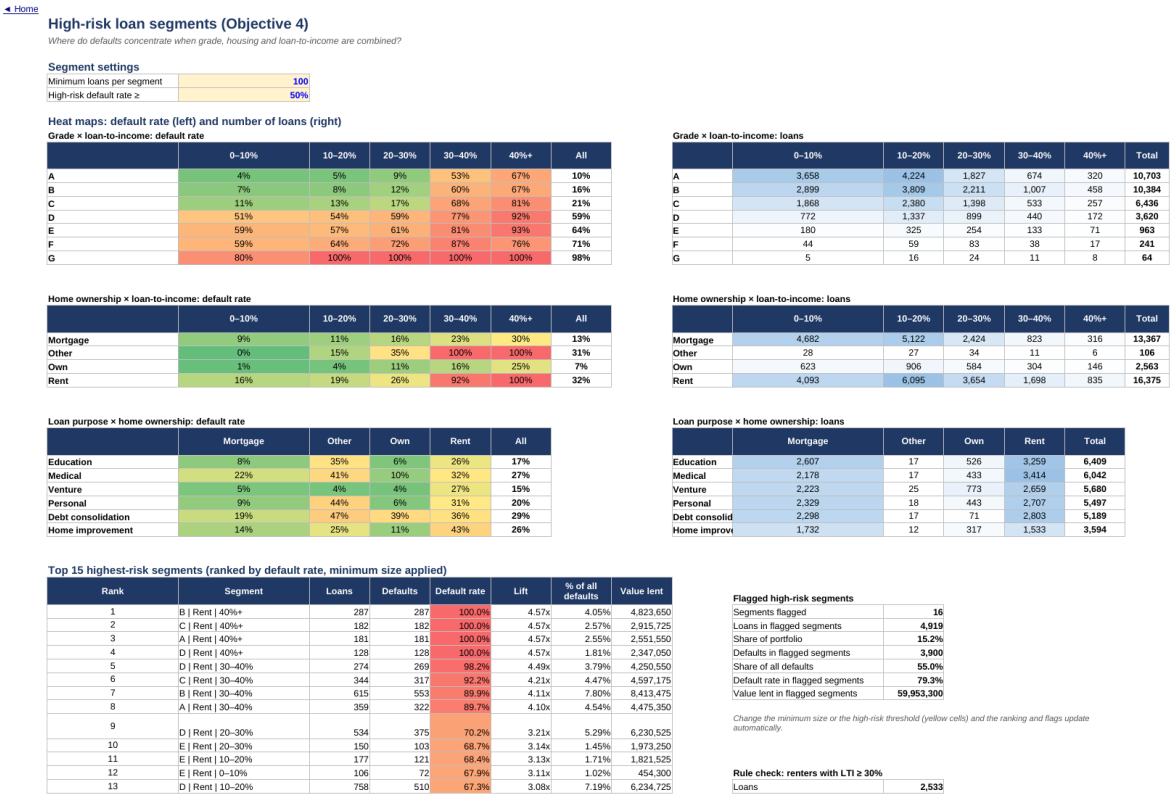
Interpretation: default rises monotonically with grade, with a sharp break between C (20.8%) and D (59.1%). Rates are priced by grade, so the overall rate–default relationship (Welch t = 57.8; default rate 9.4% under 8% vs 68.8% at 18%+) largely reflects grade. Within grades, loans priced above the median default at about the same rate as those below it (C: 20.8% vs 20.8%; D: 60.0% vs 59.0%); only grade A shows a modest gap (8.3% vs 11.9%), and F/G are too small to judge. The logistic model confirmed this: adding the rate to the grade changed test AUC by less than 0.001. **Implication:** pricing is not adding risk discrimination beyond the grade; grades D–G may be under-priced relative to 59–98% default rates. **Limitation:** 9.6% of rates are missing.

7.4 Objective 4 - High-risk loan segments

Procedure: heat maps and a 140-segment table (grade × housing × LTI band), ranked by default rate with a minimum size of 100 loans; segments at ≥ 50% default are flagged (both thresholds adjustable).

Rank	Segment (grade | housing | LTI)	Loans	Default rate	Lift	% of all defaults
1	B | Rent | 40%+	287	100.0%	4.57x	4.05%
2	C | Rent | 40%+	182	100.0%	4.57x	2.57%
3	A | Rent | 40%+	181	100.0%	4.57x	2.55%
4	D | Rent | 40%+	128	100.0%	4.57x	1.81%
5	D | Rent | 30–40%	274	98.2%	4.49x	3.79%
6	C | Rent | 30–40%	344	92.2%	4.21x	4.47%
7	B | Rent | 30–40%	615	89.9%	4.11x	7.80%
8	A | Rent | 30–40%	359	89.7%	4.10x	4.54%
9	D | Rent | 20–30%	534	70.2%	3.21x	5.29%
10	E | Rent | 20–30%	150	68.7%	3.14x	1.45%

Results: 16 segments are flagged; together they hold 4,919 loans (15.2% of the book), 55.0% of all defaults and default at 79.3%. The dominant pattern is renting combined with high LTI: renters with LTI ≥ 30% number 2,533 (7.8% of loans), default at 94.8% and account for 33.9% of all defaults, while non-renters at the same LTI default at 24.2%. Grade D–E renters are high-risk at every LTI. **Implication:** a simple, explainable rule targets a third of all defaults while touching under 8% of lending. **Limitation:** perfectly separated segments (100% default) are unusual in real data and suggest rule-generated outcomes.



◂ Home

High-risk loan segments (Objective 4)

Where do defaults concentrate when grade, housing and loan-to-income are combined?

Segment settings

Minimum loans per segment	100
High-risk default rate ≥	50%

Heat maps: default rate (left) and number of loans (right)

Grade × loan-to-income: default rate

	0–10%	10–20%	20–30%	30–40%	40%+	All
A	4%	5%	9%	53%	67%	10%
B	7%	8%	12%	60%	67%	16%
C	11%	13%	17%	68%	81%	21%
D	51%	54%	59%	77%	92%	59%
E	59%	57%	61%	81%	93%	64%
F	59%	64%	72%	87%	76%	71%
G	80%	100%	100%	100%	100%	98%

Grade × loan-to-income: loans

	0–10%	10–20%	20–30%	30–40%	40%+	Total
A	3,658	4,224	1,827	674	320	10,703
B	2,899	3,809	2,211	1,007	458	10,384
C	1,868	2,380	1,398	533	257	6,436
D	772	1,337	899	440	172	3,620
E	180	325	254	133	71	963
F	44	59	83	38	17	241
G	5	16	24	11	8	64

Home ownership × loan-to-income: default rate

	0–10%	10–20%	20–30%	30–40%	40%+	All
Mortgage	9%	11%	16%	23%	30%	13%
Other	0%	15%	35%	100%	100%	31%
Own	1%	4%	11%	16%	25%	7%
Rent	16%	19%	26%	92%	100%	32%

Home ownership × loan-to-income: loans

	0–10%	10–20%	20–30%	30–40%	40%+	Total
Mortgage	4,682	5,122	2,424	823	316	13,367
Other	28	27	34	11	6	106
Own	623	906	584	304	146	2,563
Rent	4,093	6,095	3,654	1,698	835	16,375

Loan purpose × home ownership: default rate

	Mortgage	Other	Own	Rent	All
Education	8%	35%	6%	26%	17%
Medical	22%	41%	10%	32%	27%
Venture	5%	4%	4%	27%	15%
Personal	9%	44%	6%	31%	20%
Debt consolidation	19%	47%	39%	36%	29%
Home improvement	14%	25%	11%	43%	26%

Loan purpose × home ownership: loans

	Mortgage	Other	Own	Rent	Total
Education	2,607	17	526	3,259	6,409
Medical	2,178	17	433	3,414	6,042
Venture	2,223	25	773	2,659	5,680
Personal	2,329	18	443	2,707	5,497
Debt consolid	2,298	17	71	2,803	5,189
Home improv	1,732	12	317	1,533	3,594

Top 15 highest-risk segments (ranked by default rate, minimum size applied)

Rank	Segment	Loans	Defaults	Default rate	Lift	% of all defaults	Value lent
1	B | Rent | 40%+	287	287	100.0%	4.57x	4.05%	4,823,650
2	C | Rent | 40%+	182	182	100.0%	4.57x	2.57%	2,915,725
3	A | Rent | 40%+	181	181	100.0%	4.57x	2.55%	2,551,550
4	D | Rent | 40%+	128	128	100.0%	4.57x	1.81%	2,347,050
5	D | Rent | 30–40%	274	269	98.2%	4.49x	3.79%	4,250,550
6	C | Rent | 30–40%	344	317	92.2%	4.21x	4.47%	4,597,175
7	B | Rent | 30–40%	615	553	89.9%	4.11x	7.80%	8,413,475
8	A | Rent | 30–40%	359	322	89.7%	4.10x	4.54%	4,475,350
9	D | Rent | 20–30%	534	375	70.2%	3.21x	5.29%	6,230,525
10	E | Rent | 20–30%	150	103	68.7%	3.14x	1.45%	1,973,250
11	E | Rent | 10–20%	177	121	68.4%	3.13x	1.71%	1,821,525
12	E | Rent | 0–10%	106	72	67.9%	3.11x	1.02%	454,300
13	D | Rent | 10–20%	758	510	67.3%	3.08x	7.19%	6,234,725

Flagged high-risk segments

Segments flagged	16
Loans in flagged segments	4,919
Share of portfolio	15.2%
Defaults in flagged segments	3,900
Share of all defaults	55.0%
Default rate in flagged segments	79.3%
Value lent in flagged segments	59,953,300

Change the minimum size or the high-risk threshold (yellow cells) and the ranking and flags update automatically.

Rule check: renters with LTI ≥ 30%

Loans	2,533

Figure 4. Segments sheet: heat maps and the ranked high-risk segments.

7.5 Objective 5 - Predicting default

Design

- Stratified 70/30 split (seed 42): 22,687 training and 9,724 test loans, both 21.87% default. Imputation values were frozen from the training rows only; the test set was used once, for evaluation.
- **Logistic regression:** candidates screened with a likelihood-ratio test; employment length, prior default, age, credit history and interest rate were dropped together ($\chi^2 = 10.5$, df 5, $p = 0.062$). LTI² was added after a linearity check (test AUC 0.876 → 0.880). McFadden R^2 0.373; train AUC 0.881 vs test 0.880 (no overfitting).
- **Decision tree:** depth capped at 6 for readability and Excel implementation; depth 6 / minimum leaf 50 chosen by 5-fold CV (AUC 0.902); 32 leaves; train AUC 0.906 vs test 0.905.

Key logistic results (odds ratios)

Effect	Odds ratio	Reading
Grade D vs A	20.06	Default odds 20× higher, other factors equal
Grade C vs A	2.22	The C→D step is far larger than A→C
Renter vs mortgage	2.25	More than double the odds
Owner vs mortgage	0.15	Odds cut by about 85%
Debt consolidation vs education	2.26	Highest-risk purpose
Venture vs education	0.75	Lowest-risk purpose
ln(income) +1 (≈ 2.7× income)	0.30	Higher income sharply lowers odds
Employment unknown	1.91	Missing data nearly doubles the odds
LTI, LTI²	—	Convex: modelled PD for a grade-B renter rises from 10.2% at 20% LTI to 51.6% at 40% and 86.4% at 50%

Test-set performance

Metric	Logistic	Tree	Grade D–G rule	No model
AUC (area under ROC curve)	0.880	0.905	0.728	0.500
Gini (2 × AUC − 1)	0.759	0.810	0.456	0.000

Metric	Logistic	Tree	Grade D–G rule	No model
KS statistic (max TPR − FPR)	0.625	0.701	0.357	0.000
Recall — share of defaults caught	57.5%	68.3%	43.2%	0.0%
Precision — flagged loans that defaulted	77.5%	96.4%	61.6%	0.0%
Specificity — good loans not flagged	95.3%	99.3%	92.4%	100.0%
Accuracy	87.1%	92.5%	81.7%	78.1%
F1 score	0.660	0.799	0.508	0.000
Share of loans flagged	16.2%	15.5%	15.4%	0.0%

Threshold 50% unless stated. AUC is computed in Excel on a 101-point grid; exact (rank-based) values from scikit-learn are 0.880 and 0.905.

At a 50% threshold the tree flags 1,507 test loans and catches 1,452 of 2,127 defaults (68.3%) with only 55 false alarms; the logistic model catches 1,224 with 356 false alarms. With a cost ratio of 5:1, the lowest-cost thresholds are 0.18 for the tree (recall 74.1%, 4.1% of good loans flagged) and 0.23 for the logistic model (recall 78.4%, 16.4%). Ranked by the tree, the riskiest 24% of test loans contain 77% of defaults (logistic: 20% → 66%). Both models are reasonably calibrated; the logistic model over-predicts slightly in the 20–30% PD band (24.6% predicted vs 18.6% actual).

Why the tree wins: risk here is interactive — high LTI is far more dangerous for renters, and grade D–G renters with short employment default almost always — and a single tree captures these thresholds while a main-effects logistic model cannot. **Which variables matter most:** IV ranks LTI (0.906), grade (0.860), rate (0.662), income (0.506) and housing (0.381) highest; the tree's importance is LTI 33%, grade 24%, renting 19%, income 9%, purpose 10%. Rate's high IV disappears once grade is known. As a benchmark only, a gradient-boosted model in Python reached AUC 0.944 — more accurate but not transparent or maintainable in Excel. **Limitation:** validation is on a random hold-out from the same period; IV values above 0.5 would normally prompt a leakage review.

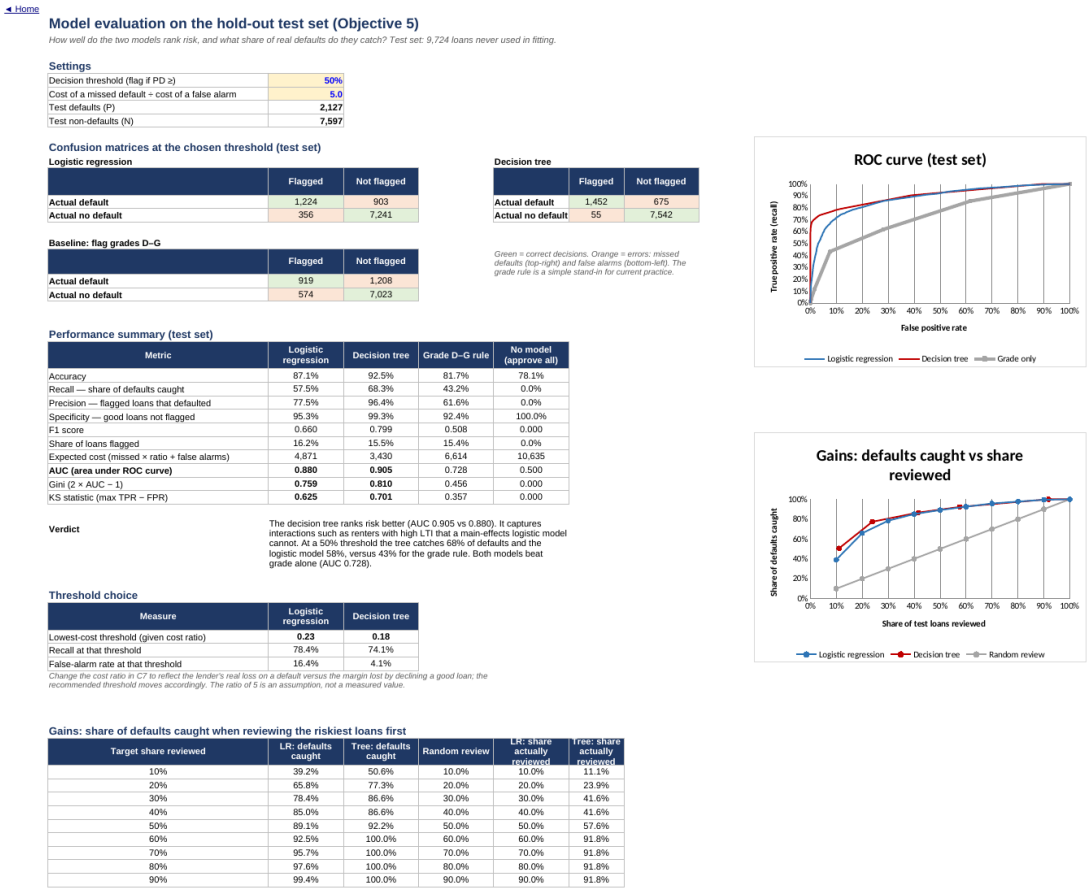


Figure 5. Model_Eval sheet: confusion matrices, performance summary, ROC and gains.

7.6 Objective 6 - Recommendations for credit-risk monitoring

Findings were translated into prioritised actions (Section 9) and a set of monitoring indicators with current values on the Insights sheet: portfolio default rate, share of lending at LTI ≥ 30%, share to renters at LTI ≥ 30%, share in grades D–G, share in flagged segments, share with unknown employment, champion-model AUC on new outcomes, and QA status.

7.7 Objective completion summary

Objective	Analytical procedure	Excel functionality	Result	Business interpretation
1	KPIs; distributions by grade, purpose, housing, amount	COUNTIFS, SUMIFS, AVERAGEIFS, RANK.EQ, defined names, data bars	32,411 loans; 21.9% default; 310,937,900 lent	High-risk book concentrated in A–C by count but with costly D–G tail
2	Band tables; χ^2, Cramér's V; Welch t; r	SUMPRODUCT, CHISQ.DIST.RT, T.DIST.2T, STDEV.P, array MEDIAN(IF)	LTI V 0.432, income 0.301, housing 0.242	Affordability and housing drive risk; age irrelevant
3	Grade table; rate bands; within-grade split	MINIFS/MAXIFS, COUNTIFS with "<="&median	C→D jump; rate adds ~nothing within grade	Grade carries the pricing signal; D–G pricing review
4	Heat maps; 140-segment ranking	LARGE + INDEX/MATCH, colour scales, input thresholds	16 segments, 55.0% of defaults	Renter + LTI ≥ 30% rule
5	Logistic regression; decision tree; evaluation; IV	SUMIFS scoring, nested IF, ROC table, LARGE(IF), Solver-ready LL	AUC 0.880 / 0.905; recall 68.3% at 50%	Tree score as second-line check
6	Findings → actions; KRIs	Live TEXT() narrative, defined names	10 findings, 8 KRIs	Monitoring framework

8. Key findings

Finding	Evidence	Interpretation	Business implication	Recommended action
Loan-to-income is the strongest driver of default	Default 21.4% at 20–30% LTI vs 64.2% at 30–40% and 74.5% above 40%; Cramér's V 0.432; IV 0.906; top tree split at ~30%	Likely explanation (not tested here): borrowers committing over 30% of income have little room to absorb shocks	Affordability does not appear to be controlled by the current grading	Introduce a 30% LTI review threshold
Renters with LTI ≥ 30% almost always default	2,533 loans, 94.8% default, 33.9% of all defaults; non-renters at same LTI 24.2%	Housing changes how dangerous high LTI is (interaction)	One rule targets a third of defaults for under 8% of lending	Decline or restructure; monitor as a KRI
Grade risk jumps between C and D	C 20.8% vs D 59.1%; G 98.4%; D–G = 15.1% of loans	Grades D–G are a distinct high-risk tier	Pricing (15–20%) is unlikely to cover 59–98% default	Review D–G pricing, limits and grade boundaries
Interest rate adds little beyond grade	Within-grade above/below-median default rates nearly equal; LR AUC unchanged with rate	Rate appears to be set from grade	Pricing is not an independent risk control	Monitor grade, LTI and housing rather than rate
Unknown employment is a risk signal	31.7% default vs 21.9% overall; odds ratio 1.91 in the logistic model	Incomplete applications may indicate weaker borrowers or weaker checks	Data gaps hide risk	Make employment a mandatory verified field
Default is highly predictable from application data	Tree AUC 0.905, logistic 0.880, grade 0.728	A small set of application fields separates good from bad borrowers	Better targeting than grade alone	Pilot the tree score as a second-line check

Findings describe associations in historical, approved-only data; they are not causal claims.

9. Business recommendations

Priority	Recommendation	Supporting evidence	Assumptions / dependencies
Immediate	Set an LTI review threshold at 30%; above 40% require a lower amount or extra security	Default 21% → 64% across the 30% line	Thresholds should be confirmed on real recent data
Immediate	Make "renter + LTI ≥ 30%" a hard decline/restructure rule and a KRI	94.8% default; 33.9% of defaults	Fair-lending review of using housing status
Immediate	Make employment length mandatory and verified	Unknown employment defaults at 31.7%	Process change in origination
Immediate	Adopt the dashboard and Insights KRIs in monthly risk reporting	Live, reconciled measures	Data refresh process (Power Query script provided)
Medium term	Pilot the decision-tree score as a second-line check; route High/Very-high applications to senior review	Very-high band 96.7% default; AUC 0.905	Validate on recent applications before use; governance sign-off
Medium term	Agree the cost of a missed default vs a declined good loan; set the threshold accordingly	Lowest-cost threshold tool on Model_Eval	Needs loss and margin data from Finance
Medium term	Review pricing and limits for grades D–G; recalibrate grade boundaries	C→D jump; rate adds nothing within grade	Needs loss-given-default to price properly
Long term	Capture origination date, loan ID, balance at default, recoveries, written default definition; fix LTI calculation at source	Gaps listed in Section 10	Systems change
Long term	Move to a governed modelling platform for more accurate models with monitoring (drift, stability)	Benchmark AUC 0.944 vs 0.905	Model-risk governance capability

10. Technical and analytical limitations

Observed limitations (in the data)

- No identifiers, dates, loss amounts or currency; default definition and observation window undocumented.
- Missing interest rate (9.6%) and employment length (2.7%); 11 placeholder rates; 7 impossible values.
- 428 defaulted records with an inconsistent loan-to-income field — a sign the outcome or amounts may have been edited after the fact, or of how the data was generated.
- Near-deterministic segments (e.g. 100% default for some renter/LTI groups) suggest rule-generated data; real-world accuracy would be lower.

Methodological limitations

- Random hold-out validation only; no out-of-time or out-of-population test.
- Logistic model has main effects only (plus LTI^2); interactions are left to the tree.
- Tree PDs are coarse (32 distinct values); ties affect gains and threshold granularity.
- Median imputation for tree inputs; frozen at fit time.
- AUC in Excel is a 101-point trapezoid approximation (differs from exact by < 0.001).
- Chi-square p-values are uninformative at this sample size; effect sizes are used instead.

Contextual limitations

- Only approved loans — no information on rejected applicants (no reject inference); models may not generalise to the full applicant population.
- No macroeconomic context, product terms, collateral, or repayment history.
- Cost of errors unknown; the 5:1 cost ratio is an assumption for illustration.

Assumptions

- `loan_status = 1` means default; grade A is the lowest-risk grade; `cb_` fields come from a credit bureau; all amounts share one currency; fields are as at application.
- Exact duplicates are duplicated records, not distinct identical loans.

Additional data that would most improve the analysis

Data	Why
Origination and default dates	Trend, vintage and seasonality analysis; out-of-time validation
Loan ID and borrower ID	Deduplication certainty; repeat-borrower behaviour
Balance at default, recoveries, term	Loss given default, exposure at default and expected loss; pricing review
Declined applications	Reject inference and fair evaluation of new rules
Written default definition	Consistent KPI definition and model target

11. Validation and quality assurance

11.1 Data validation

- Raw (32,581) = cleaned (32,411) + excluded (170); exclusions listed record by record.
- Duplicate and invalid-value counts confirmed independently; missing-value categories reconcile (889 unknown employment, 3,105 unrecorded rates).
- Grade, purpose, LTI and segment tables reconcile to portfolio totals; train + test = all loans with identical default rates.

11.2 Analysis validation

Every headline KPI, all eleven chi-square statistics and Cramér's V values, all eight Welch t-statistics and point-biserial correlations, the seven band assignments for every loan, all 32,411 logistic PDs (max difference < 1e-9), all tree-leaf assignments and PDs, AUCs, recall, precision, gains and segment flags were reconciled against independent calculations in pandas, SciPy, statsmodels and scikit-learn. All matched.

11.3 Deliverable validation — what was and was not tested

Item	Status
Workbook formulas	All 620,601 formulas recalculated in LibreOffice Calc 24.2 with zero errors; the delivered file re-opened and recalculated cleanly. The workbook was not opened in Microsoft Excel in the build environment.
Excel compatibility measures	Excel-2010+ function names prefixed correctly; no XLOOKUP or spilling functions; table column names and calculated-column formulas set explicitly; workbook flagged to recalculate fully on open
QA_Checks sheet	28 / 28 checks PASS
Charts and layout	Reviewed visually from LibreOffice exports; rendering in Excel may differ slightly
Power Query script	Written to mirror the pandas rules; not executed (no Excel in the build environment)
Python script	Executed end to end; regenerates the processed data and model parameters
Report and README	Figures generated from the recalculated workbook

11.4 QA checklist

#	Check	Status
1	Raw records in Data_Raw	PASS
2	Cleaned + excluded records = raw records	PASS
3	Duplicate records excluded	PASS
4	Invalid-age records excluded	PASS
5	Loan IDs strictly increasing (so no duplicates)	PASS
6	No blank grades, purposes, housing or outcomes	PASS
7	Missing employment length = Unknown band	PASS
8	Missing interest rate = Not recorded band	PASS
9	No ages above 100 remain	PASS
10	Defaults (cleaned)	PASS
11	Total value lent (cleaned)	PASS
12	Grade table loans reconcile to portfolio	PASS
13	Grade table value reconciles to portfolio	PASS
14	Purpose table defaults reconcile	PASS
15	LTI band loans reconcile	PASS
16	Segment table covers every loan	PASS
17	LTI mismatch flags (all on defaults)	PASS
18	Train + test = all loans	PASS

#	Check	Status
19	Train default rate ≈ test default rate (stratified split)	PASS
20	Every loan routed to a tree leaf	PASS
21	Logistic PDs all between 0 and 1	PASS
22	Logistic: mean predicted PD = actual rate on train (MLE property)	PASS
23	Logistic log-likelihood matches statsmodels	PASS
24	Logistic test AUC matches scikit-learn (grid approximation)	PASS
25	Tree test AUC matches scikit-learn (grid approximation)	PASS
26	Confusion matrices sum to test loans	PASS
27	Frozen employment imputation = live training median	PASS
28	Frozen rate imputation = live training medians (sum of 7 grades)	PASS

Recommended first step after download: open the workbook in Excel, confirm the Home sheet shows "ALL PASS", and try the dashboard filters.

12. Lessons and professional competencies demonstrated

Competency	Where it is demonstrated
Business problem framing	Problem statement, objective-to-analysis mapping, decisions supported (Sections 1–2)
Data quality and preparation	Systematic assessment; duplicate-block detection; LTI integrity investigation and sensitivity check; full cleaning log and record reconciliation
Advanced Excel	Structured tables and calculated columns; COUNTIFS/SUMIFS/AVERAGEIFS/MINIFS/MAXIFS; INDEX/MATCH; array formulas; SUMPRODUCT statistics; data validation; conditional formatting; defined names; charts; Solver-ready model
Statistical reasoning	Chi-square with effect sizes, Welch t-tests, assumption checks, within-grade confounding test, linearity check, likelihood-ratio variable selection
Predictive modelling and validation	Train/test design, cross-validated tuning, baseline comparison, ROC/AUC, KS, calibration, gains, cost-based threshold
Credit-risk domain knowledge	PD, risk bands, WoE/IV, Gini, grade calibration, affordability (LTI), reject-inference and loss limitations
Tool design and usability	Interactive dashboard, applicant scorer, input validation, settings sheet, QA sheet, documentation inside the workbook
Communication	Live findings, prioritised recommendations, README for non-technical readers
Professional integrity	Transparent disclosure of supporting tools, untested components, data provenance and limitations

Lessons

- The biggest insight came from an interaction (housing × LTI), not a single variable — always look at combinations once main effects are known.
- A strong univariate signal (interest rate, IV 0.66) can be redundant; testing within strata prevents over-claiming.
- Data-integrity anomalies concentrated in one outcome class deserve investigation before modelling — they can inflate apparent accuracy.
- Transparency has a price: the Excel-native tree trails a gradient-boosted benchmark by about 0.04 AUC, a trade-off worth stating explicitly to stakeholders.

13. Conclusion

The project met all six objectives. It established that 21.9% of 32,411 loans defaulted, showed that loan-to-income, grade, income and housing drive that risk while age and credit history do not, demonstrated that interest rate adds little beyond grade, and located a small set of segments - led by renters borrowing 30% or more of income - that account for over half of all defaults. Two transparent models estimate default probability for every loan; the decision tree (AUC 0.905) clearly outperforms both the logistic regression (0.880) and the existing grade (0.728).

These results are delivered as a working Excel tool rather than a static report: management can monitor any part of the book, adjust risk thresholds, compare models, score new applications and check data quality in one place. The recommendations include the data improvements needed before the models are used for real lending decisions.

Appendix A. Logistic regression coefficients

Variable	Level	Coefficient	Std error	z	p-value	Odds ratio
(intercept)	—	9.4018	0.5194	18.10	< 0.0001	1.21e+4
loan_grade	A	0.0000	ref.			1.000
loan_grade	B	0.3928	0.0591	6.65	< 0.0001	1.481
loan_grade	C	0.7974	0.0631	12.64	< 0.0001	2.220
loan_grade	D	2.9986	0.0689	43.53	< 0.0001	20.057
loan_grade	E	3.3149	0.1086	30.52	< 0.0001	27.519
loan_grade	F	3.8462	0.2122	18.12	< 0.0001	46.817
loan_grade	G	7.2419	1.0627	6.81	< 0.0001	1.40e+3
person_home_ownership	MORTGAGE	0.0000	ref.			1.000
person_home_ownership	OTHER	0.6639	0.3542	1.87	0.0609	1.942
person_home_ownership	OWN	-1.9086	0.1269	-15.04	< 0.0001	0.148
person_home_ownership	RENT	0.8128	0.0486	16.73	< 0.0001	2.254
loan_intent	EDUCATION	0.0000	ref.			1.000
loan_intent	DEBTCONSOLIDATION	0.8161	0.0692	11.79	< 0.0001	2.262
loan_intent	HOMEIMPROVEMENT	0.9439	0.0790	11.94	< 0.0001	2.570
loan_intent	MEDICAL	0.6206	0.0674	9.21	< 0.0001	1.860
loan_intent	PERSONAL	0.1424	0.0726	1.96	0.0497	1.153
loan_intent	VENTURE	-0.2852	0.0763	-3.74	0.0002	0.752
lti	—	-2.1519	0.7501	-2.87	0.0041	0.116
lti_squared	—	22.2130	1.6026	13.86	< 0.0001	4.44e+9
ln_income	—	-1.2126	0.0460	-26.37	< 0.0001	0.297
emp_unknown	—	0.6473	0.1200	5.40	< 0.0001	1.910

Training n = 22,687; log-likelihood -7469.7; McFadden R^2 0.373. Odds ratios for LTI and LTI² are per unit of LTI (i.e. per 100% of income) and are not meaningful individually; read them through the PD-by-LTI table on Model_LR.

Appendix B. Decision tree rules

Leaf	Rule	Train loans	Train default	Leaf PD	Test loans	Test default
L01	LTI ≤ 15.5%; Grade A–C; Income ≤ 19,996; Rate ≤ 10.83%; Age ≤ 27.5	51	5.9%	7.5%	20	15.0%
L02	LTI ≤ 15.5%; Grade A–C; Income ≤ 19,996; Rate > 10.83%; Age ≤ 27.5	50	14.0%	15.4%	28	21.4%
L03	LTI ≤ 15.5%; Grade A–C; Income ≤ 19,996; Age > 27.5	54	31.5%	32.1%	24	25.0%
L04	LTI 15.5%–30.4%; Grade A–C; Income ≤ 19,996	255	100.0%	99.6%	111	100.0%
L05	LTI ≤ 30.4%; Grade A–C; Income 19,996–33,918; Purpose not Home improvement	1,949	11.0%	11.0%	790	9.9%
L06	LTI ≤ 30.4%; Grade A–C; Income 33,918–34,048; Purpose not Home improvement	110	39.1%	39.3%	39	28.2%
L07	LTI ≤ 30.4%; Grade A–C; Income 19,996–34,048; Home not Rent; Purpose: Home improvement	90	23.3%	23.9%	42	16.7%
L08	LTI ≤ 30.4%; Grade A–C; Income 19,996–34,048; Home: Rent; Purpose: Home improvement	56	100.0%	98.3%	26	100.0%
L09	LTI ≤ 30.4%; Grade A–C; Income > 34,048; Purpose not Venture; Rate ≤ 11.18%	7,752	4.8%	4.8%	3,330	5.0%
L10	LTI ≤ 30.4%; Grade A–C; Income > 34,048; Purpose: Venture; Rate ≤ 11.18%	1,703	0.0%	0.1%	705	0.0%
L11	LTI ≤ 30.4%; Grade A–C; Income 34,048–79,994; Rate > 11.18%	3,440	11.0%	11.1%	1,473	10.8%
L12	LTI ≤ 30.4%; Grade A–C; Income > 79,994; Rate > 11.18%	1,640	5.0%	5.1%	741	5.3%
L13	LTI ≤ 30.4%; Grade E–G; Home not Rent; Purpose not Medical/Education; Employment ≤ 2.5 yrs	219	32.4%	32.6%	92	35.9%
L14	LTI ≤ 30.4%; Grade E–G; Home not Rent; Purpose: Education; Employment ≤ 2.5 yrs	77	13.0%	13.9%	35	20.0%
L15	LTI ≤ 30.4%; Grade E–G; Home not Rent; Purpose: Medical; Employment ≤ 2.5 yrs	54	88.9%	87.5%	20	80.0%
L16	LTI ≤ 30.4%; Grade E–G; Home: Rent; Employment ≤ 2.5 yrs	671	100.0%	99.9%	288	100.0%
L17	LTI ≤ 30.4%; Grade E–G; Income ≤ 34,900; Purpose not Debt consolidation/Medical; Employment > 2.5 yrs	128	45.3%	45.4%	74	44.6%
L18	LTI ≤ 30.4%; Grade E–G; Income > 34,900; Purpose not Debt consolidation/Medical; Employment > 2.5 yrs	1,063	17.2%	17.3%	434	15.9%
L19	LTI ≤ 30.4%; Grade E–G; Home not Mortgage; Purpose: Medical; Employment > 2.5 yrs	212	77.4%	77.1%	91	74.7%
L20	LTI ≤ 30.4%; Grade E–G; Home: Mortgage; Purpose: Medical; Employment > 2.5 yrs	109	100.0%	99.1%	61	100.0%
L21	LTI ≤ 30.4%; Grade E–G; Home not Mortgage; Purpose: Debt consolidation; Employment > 2.5 yrs; Rate ≤ 15.08%	51	76.5%	75.5%	19	89.5%
L22	LTI ≤ 30.4%; Grade E–G; Home not Mortgage; Purpose: Debt consolidation; Employment > 2.5 yrs; Rate > 15.08%	105	90.5%	89.7%	50	96.0%
L23	LTI ≤ 30.4%; Grade E–G; Home: Mortgage; Purpose: Debt consolidation; Employment > 2.5 yrs	140	100.0%	99.3%	58	100.0%
L24	LTI > 30.4%; Income ≤ 19,900; Home not Rent	63	100.0%	98.5%	23	100.0%
L25	LTI > 30.4%; Grade A–B; Income > 19,900; Home not Rent/Own	467	12.0%	12.2%	187	13.4%
L26	LTI > 30.4%; Grade D–C; Income > 19,900; Home not Rent/Own	114	29.8%	30.2%	66	25.8%
L27	LTI > 30.4%; Grade A–C; Income > 19,900; Home: Own	209	0.0%	0.5%	92	0.0%
L28	LTI > 30.4%; Grade E–G; Income > 19,900; Home not Rent; Purpose not Medical; Age ≤ 24.5	61	57.4%	57.1%	25	32.0%
L29	LTI > 30.4%; Grade E–G; Income > 19,900; Home not Rent; Purpose not Medical; Age > 24.5	86	38.4%	38.6%	45	37.8%
L30	LTI > 30.4%; Grade E–G; Income > 19,900; Home not Rent; Purpose: Medical	50	96.0%	94.2%	23	78.3%

Leaf	Rule	Train loans	Train default	Leaf PD	Test loans	Test default
L31	LTI 30.4%–30.8%; Home: Rent	55	89.1%	87.7%	34	94.1%
L32	LTI > 30.8%; Home: Rent	1,603	100.0%	99.9%	678	100.0%